

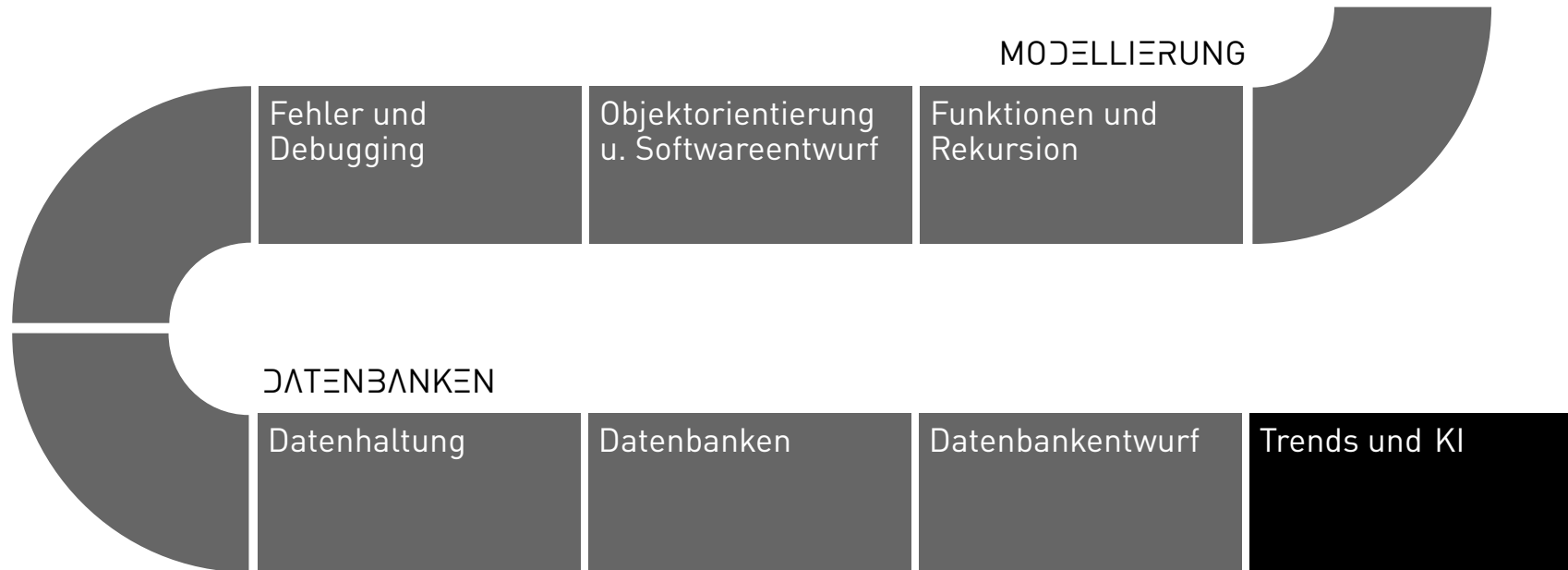
Programming and Databases

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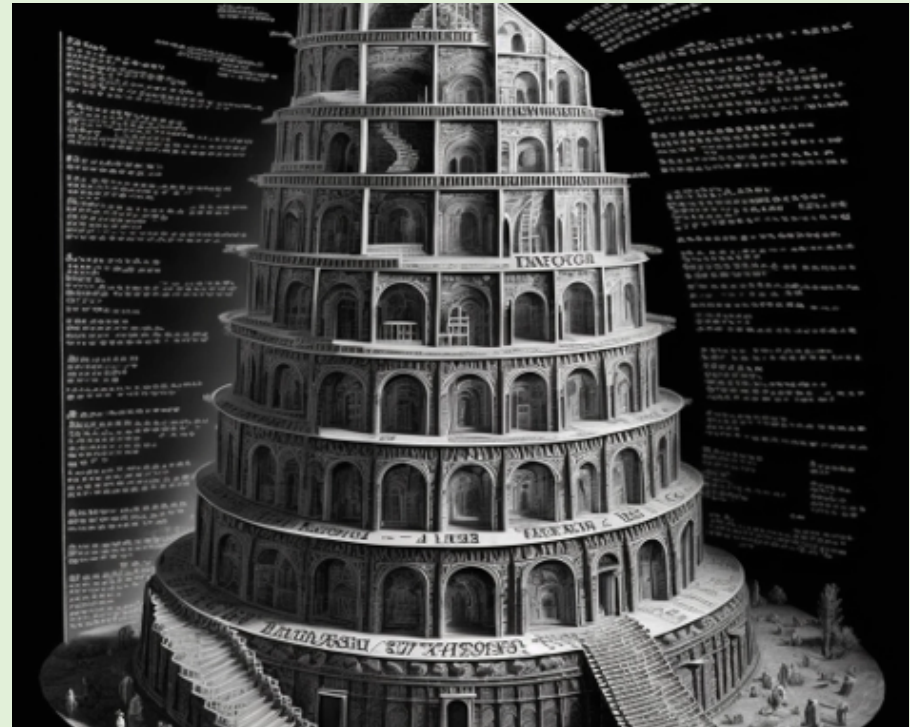
Creating Tables

MIDJOURNEY: THE CREATION OF SQL, REF. MICHELANGELO

PROCEDURE



CREATING TABLES WITH SQL



Midjourney: Construction plan of the tower of babel

FINAL STEP: IMPLEMENTATION OF THE ER DIAGRAM

Mapping the ER model to SQL commands to create the tables

- Create the tables
- For every attribute of every relation, find a sensible data type
- Determine whether attributes are nullable or non-nullable
- Define the primary key
- Define foreign keys
- Declare additional indexes to optimize performance
- Define access rights

To create the table, you need the **CREATE TABLE** SQL command.

This does not insert any data yet, but only creates an empty scaffold.

DDL - DATA DEFINITION LANGUAGE

Allows operations on the definitions of data structures:

- **CREATE** – creating tables
- **ALTER** – modifying tables
- **DROP** – deleting tables
- **TRUNC** – truncating data but not deleting the table

The possible data structures are:

- **TABLE** – tables for storing data
- **VIEW** – dynamically generated views on tables
- **MATERIALIZED VIEW** – tables that are derived from other tables

CREATING TABLES IN SQL

```
CREATE TABLE table_name (  
    attribute1_name attribute1_type attribute1_constraints,  
    attribute2_name attribute2_type attribute2_constraints,  
    .../  
    table_constraints  
)
```

DATA TYPES IN SQLITE VS. PYTHON

SQLite	Python	Description
Boolean (INTEGER)	bool	Boolean values
INTEGER	int	Integer
REAL	float	Floating-point number
NUMERIC	-	Interprets arbitrary input data as numbers
TEXT	str	Text string, stored in UTF-8 or UTF-16
BLOB	bytes	Arbitrary block of binary data
NULL	None	No data

EXAMPLE: CREATING TABLES IN SQLITE

For our geometry example, the following CREATE TABLE statements are produced:

```
CREATE TABLE Points (  
    point_id INTEGER,  
    x REAL,  
    y REAL  
);
```

```
CREATE TABLE Lines (  
    lines_id INTEGER,  
    start INTEGER,  
    end INTEGER  
);
```

KEYS AND CONSTRAINTS IN SQL

- Primary and secondary keys are usually represented by INTEGER columns.
- The primary key of every table must be explicitly marked by the PRIMARY KEY constraint.
- Other constraints that must be observed when inserting new data:
- **NOT NULL** – Values must not be null
- **AUTOINCREMENT** – Numeric primary keys are generated automatically
- **UNIQUE** – Values must be unique; duplicates are not allowed
- **CHECKED** – Additional logical constraint for new values
- **FOREIGN KEY** – Value of an existing foreign key

EXAMPLE: PRIMARY KEYS, FOREIGN KEYS, AND CONSTRAINTS

```
CREATE TABLE Points (  
    point_id INTEGER PRIMARY KEY AUTOINCREMENT,  
    x REAL NOT NULL,  
    y REAL NOT NULL  
);  
  
CREATE TABLE Lines (  
    lines_id INTEGER PRIMARY KEY AUTOINCREMENT,  
    start INTEGER NOT NULL,  
    end INTEGER NOT NULL,  
    FOREIGN KEY(start) REFERENCES Points(point_id),  
    FOREIGN KEY(end) REFERENCES Points(point_id)  
);
```

CREATE TABLE ADVANCED FEATURES

Additional options:

- **DEFAULT** : Set the default value for new elements
- **WITHOUT ROWID** : Tables without a ROWID
- **STRICT** : Enforces strict data types for the table

ADDING DATA TO TABLES

Data is added to tables using the SQL command **INSERT**

Add data in the order of the column names:

```
INSERT INTO Points VALUES (1,  
54.083336, 12.108811);  
INSERT INTO Points VALUES (2,  
12.094167, 54.075211);  
INSERT INTO Lines VALUES (1, 1, 2);
```

or with explicit column names:

```
INSERT INTO Points (x, y)  
VALUES (54.083336, 12.108811);  
INSERT INTO Points (x, y)  
VALUES (12.094167, 54.075211);  
INSERT INTO Lines (start, end)  
VALUES (1, 2);
```

CREATE A TABLE FROM EXISTING DATA

```
CREATE TABLE tablename AS  
SELECT ... FROM ... WHERE ...
```

It has no primary key; it can only be defined afterwards.

ALTER TABLE - STATEMENT

Renaming tables and columns, and adding and dropping columns

```
ALTER TABLE tablename TO newtablename
```

Example:

```
ALTER TABLE Points RENAME TO Punkte
```

ALTER TABLE - EXAMPLE

Rename owner to customer, add a column for account number

```
ALTER TABLE tablename ADD COLUMN column_name column_type
```

Example:

```
ALTER TABLE PolygonType ADD COLUMN maxPoints INTEGER
```

DROP TABLE - STATEMENT

Deleting entire tables.

```
DROP TABLE tablename
```

Example:

```
DROP TABLE Points
```

Questions?

